МИНИСТЕРСТВО ЦИФРОВОГО РАЗВИТИЯ И МАССОВЫХ КОММУНИКАЦИЙ

Ордена Трудового Красного Знамени
Федеральное государственное бюджетное общеобразовательное учреждение высшего образования

“Московский технический университет связи и информатики”
(МТУСИ)
Кафедра “Программная инженерия”

Лабораторная работа №5

по дисциплине “Введение в информационные технологии”

Выполнил: Студент группы
БПИ2503
Яричевский Даниил

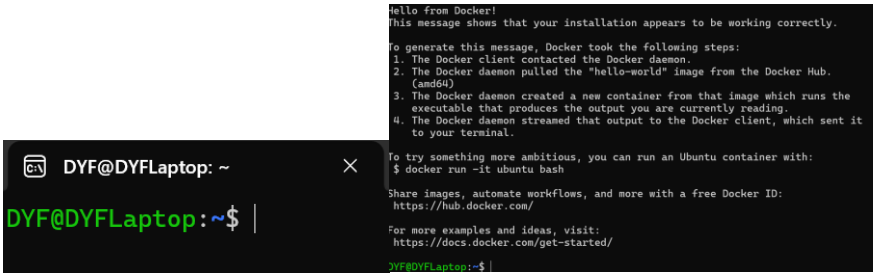
Москва
2025

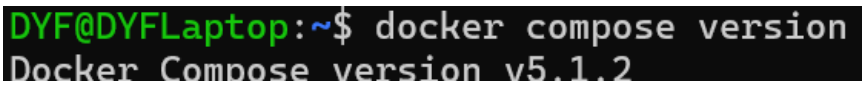
Цель работы

Изучить основы docker и деплой проекта.

Ход работы

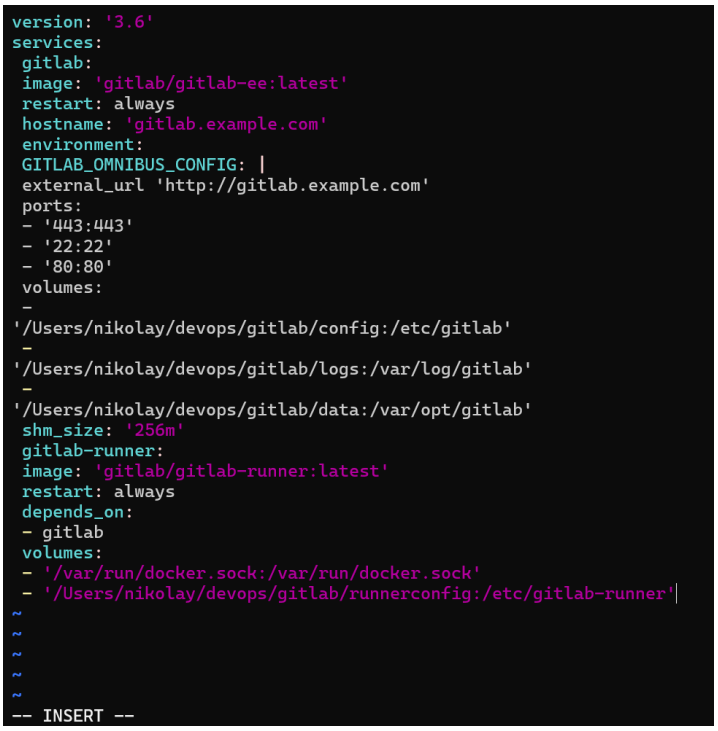
1. Установим Linux с помощью wsl, docker и docker compose

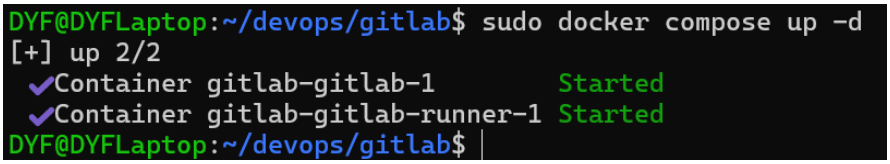




2. Установим gitlab



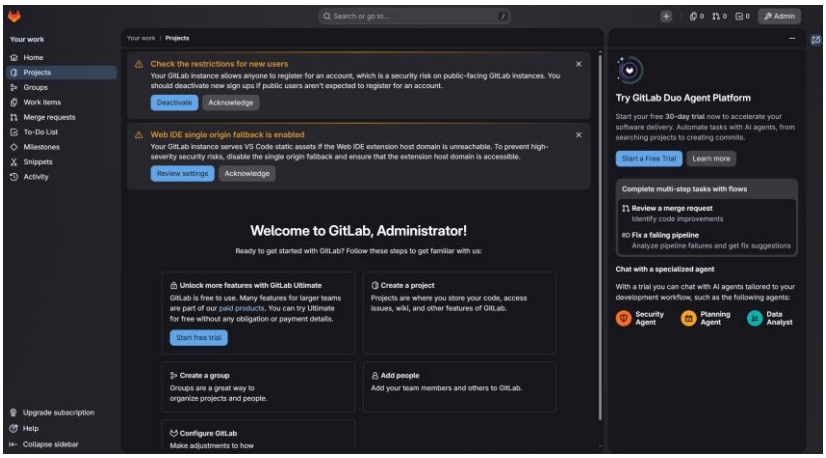




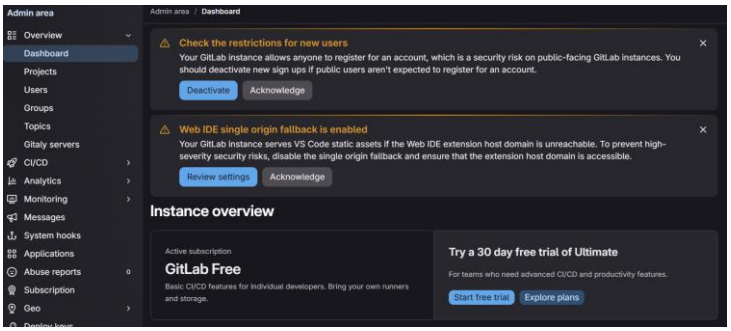
```
DYF@DYFLaptop:~/devops/gitlab$ ls
config  data  docker-compose.yaml  logs  runner-config  runnerconfig
DYF@DYFLaptop:~/devops/gitlab$ cd config
DYF@DYFLaptop:~/devops/gitlab/config$ ls
gitlab.rb           initial_root_password  ssh_host_ecdsa_key.pub  ssh_host_ed25519_key.pub  ssh_host_rsa_key.pub
gitlab-secrets.json  ssh_host_ecdsa_key     ssh_host_ed25519_key    ssh_host_rsa_key          trusted-certs
DYF@DYFLaptop:~/devops/gitlab/config$ cat initial root password
cat: initial: No such file or directory
cat: root: No such file or directory
cat: password: No such file or directory
DYF@DYFLaptop:~/devops/gitlab/config$ cat initial_root_password
cat: initial_root_password: Permission denied
DYF@DYFLaptop:~/devops/gitlab/config$ sudo cat initial_root_password
# WARNING: This password is only valid if ALL of the following are true:
#          • You set it manually via the GITLAB_ROOT_PASSWORD environment variable
#            OR the gitlab_rails['initial_root_password'] setting in /etc/gitlab/gitlab.rb
#          • You set it BEFORE the initial database setup (typically during first installation)
#          • You have NOT changed the password since then (via web UI or command line)
#
#          If this password doesn't work, reset the admin password using:
#          https://docs.gitlab.com/security/reset_user_password/#reset-the-root-password

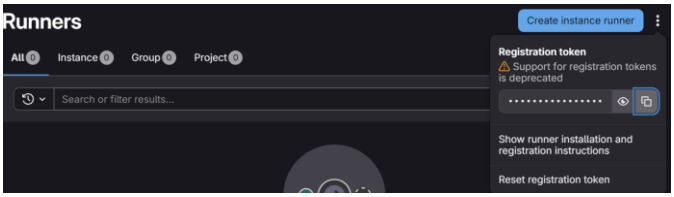
Password: jeGQKilxLdJAN/JjiNLVNZCxveDTNQlMA27J8APik3Y=

# NOTE: This file is automatically deleted after 24 hours on the next reconfigure run.
DYF@DYFLaptop:~/devops/gitlab/config$
```



3. Войдем с админ правами



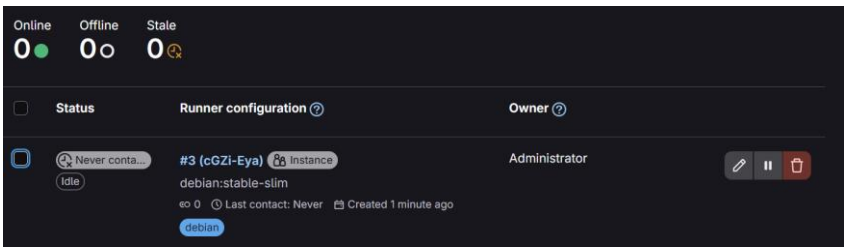



```
DYF@DYFLaptop:~/devops/gitlab/config$ sudo docker run --rm -it   -v "$PWD/runnerconfig:/etc/gitlab-runner"   --network="
gitlab_default"   gitlab/gitlab-runner register
Runtime platform                                    arch=amd64 os=linux pid=8 revision=ad1797b3 version=18.11.3
Running in system-mode.

Enter the GitLab instance URL (for example, https://gitlab.com/):
http://gitlab.example.com/
Enter the registration token:
H41baxYJqXuEKuDW_s4d
Enter a description for the runner:
[7103576188f0]: debian:stable-slim
Enter tags for the runner (comma-separated):
debian
Enter optional maintenance note for the runner:

WARNING: Support for registration tokens and runner parameters in the 'register' command has been deprecated in GitLab R
unner 15.6 and will be replaced with support for authentication tokens. For more information, see https://docs.gitlab.co
m/ci/runners/new_creation_workflow/
Registering runner... succeeded                     correlation_id=01KS8KF2MFW5R57CV6XNA846E1 runner=H41baxYJq runner_na
me=debian:stable-slim
Enter an executor: docker-autoscaler, ssh, virtualbox, docker-windows, docker+machine, kubernetes, parallels, shell, cus
tom, instance, docker:
docker
Enter the default Docker image (for example, ruby:3.3):
debian:stable-slim
Runner registered successfully. Feel free to start it, but if it's running already the config should be automatically re
loaded!

Configuration (with the authentication token) was saved in "/etc/gitlab-runner/config.toml"
DYF@DYFLaptop:~/devops/gitlab/config$
```

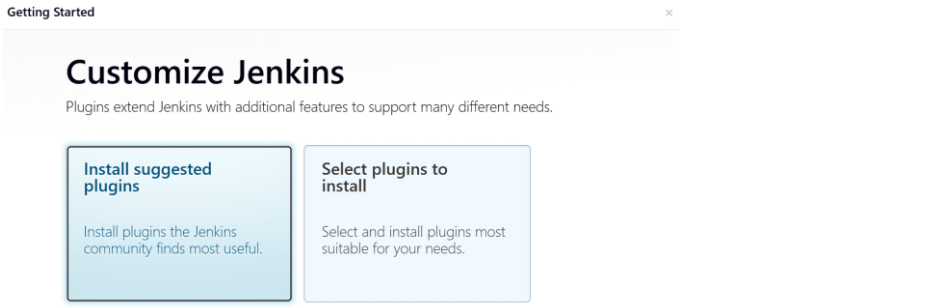


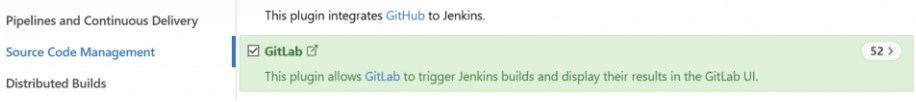
4. Установим Jenkins

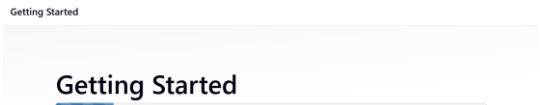
```
DYF@DYFLaptop:~/devops/gitlab$ sudo docker compose stop
[+] stop 2/2
 ✔Container gitlab-gitlab-runner-1 Stopped
 ✔Container gitlab-gitlab-1        Stopped
DYF@DYFLaptop:~/devops/gitlab$ cd ..
DYF@DYFLaptop:~/devops$ mkdir jenkins
DYF@DYFLaptop:~/devops$ cd jenkins
DYF@DYFLaptop:~/devops/jenkins$ mkdir data
DYF@DYFLaptop:~/devops/jenkins$ mkdir certs
DYF@DYFLaptop:~/devops/jenkins$ vim docer-compose.yaml
DYF@DYFLaptop:~/devops/jenkins$
```

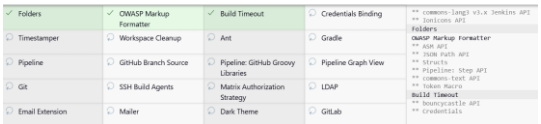
```
DYF@DYFLaptop:~/devops/jenkins$ sudo docker compose up -d
[+] up 34/34
 ✔Image docker:dind                Pulled
 ✔Image jenkins/jenkins:lts-jdk17 Pulled
 ✔Container jenkins-jenkins-1      Started
 ✔Container jenkins-docker-1       Started
DYF@DYFLaptop:~/devops/jenkins$
```

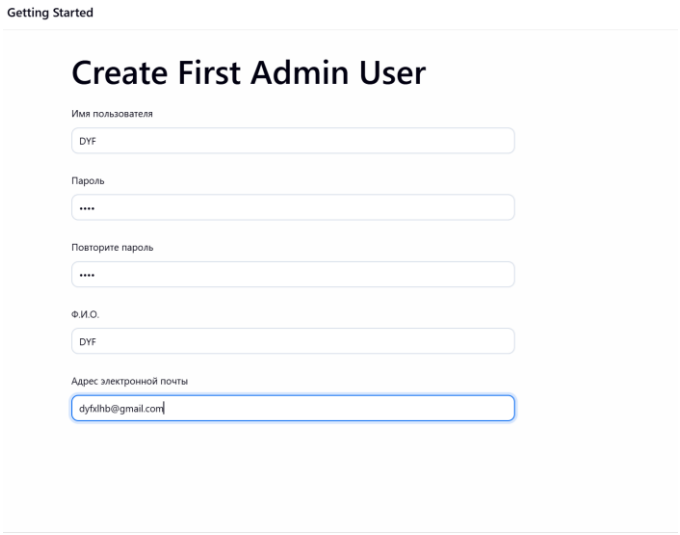
```
DYF@DYFLaptop:~/devops/jenkins$ cd jenkins_data
DYF@DYFLaptop:~/devops/jenkins/jenkins_data$ ls
config.xml                    jenkins.telemetry.Correlator.xml  plugins                 secrets      users
copy_reference_file.log       jobs                              secret.key              updates      war
hudson.model.UpdateCenter.xml  nodeMonitors.xml                 secret.key.not-so-secret  userContent
DYF@DYFLaptop:~/devops/jenkins/jenkins_data$ cd secrets
DYF@DYFLaptop:~/devops/jenkins/jenkins_data/secrets$ ls
hudson.model.User.DIRNAMES  initialAdminPassword  jenkins.model.Jenkins.crumbSalt  master.key
DYF@DYFLaptop:~/devops/jenkins/jenkins_data/secrets$ cat initialAdminPassword
211d0b43a7114626b1002c4dfd3e6844
DYF@DYFLaptop:~/devops/jenkins/jenkins_data/secrets$
```

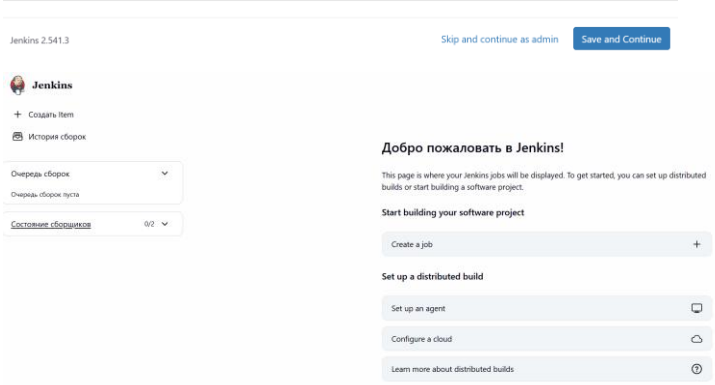












Вывод

Изучил основы docker